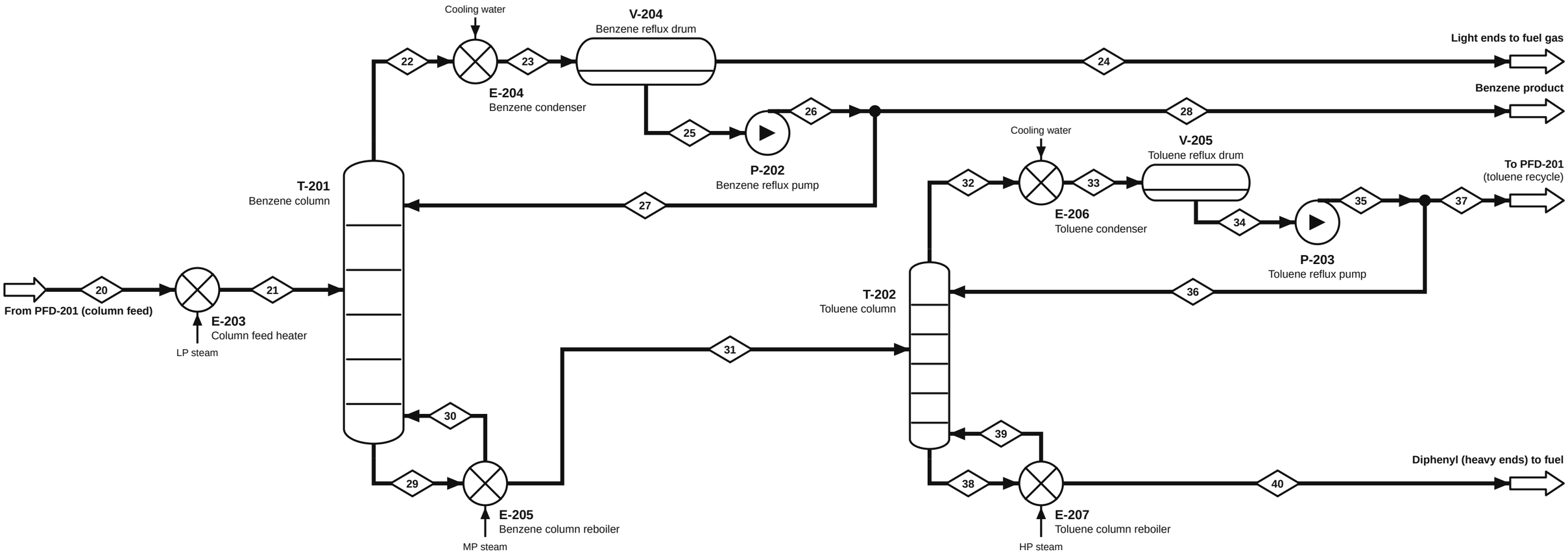
PROCESS FLOW DIAGRAM — PFD-202

Toluene hydrodealkylation to benzene | Separation section | Sheet 2 of 2



Stream table: sample values (flows from a simple component-balance model; temperatures and pressures nominal, column temperatures near ideal bubble points)

Stream	Phase	Temperature (°C)	Pressure (kPa)	Total (kmol/h)	Hydrogen (kmol/h)	Methane (kmol/h)	Benzene (kmol/h)	Toluene (kmol/h)	Diphenyl (kmol/h)
20	Liquid	37	280	130.2	0.1	1.2	95.9	32.5	0.5
21	Liquid	90	250	130.2	0.1	1.2	95.9	32.5	0.5
22	Vapour	108	220	211.5	0.1	1.2	209.8	0.4	0.0
23	Mixed	45	200	211.5	0.1	1.2	209.8	0.4	0.0
24	Vapour	45	200	1.4	0.1	1.2	0.1	0.0	0.0
25	Liquid	45	200	210.1	0.0	0.0	209.7	0.4	0.0
26	Liquid	45	400	210.1	0.0	0.0	209.7	0.4	0.0
27	Liquid	45	400	114.6	0.0	0.0	114.4	0.2	0.0
28	Liquid	45	400	95.5	0.0	0.0	95.3	0.2	0.0
29	Liquid	141	230	244.9	0.0	0.0	8.4	235.8	0.7
30	Vapour	142	230	211.6	0.0	0.0	7.9	203.5	0.2
31	Liquid	142	230	33.3	0.0	0.0	0.5	32.3	0.5
32	Vapour	127	160	58.5	0.0	0.0	0.9	57.6	0.0
33	Liquid	120	150	58.5	0.0	0.0	0.9	57.6	0.0
34	Liquid	120	150	58.5	0.0	0.0	0.9	57.6	0.0
35	Liquid	121	350	58.5	0.0	0.0	0.9	57.6	0.0
36	Liquid	121	350	26.0	0.0	0.0	0.4	25.6	0.0
37	Liquid	121	350	32.5	0.0	0.0	0.5	32.0	0.0
38	Liquid	134	170	59.2	0.0	0.0	0.0	52.9	6.3
39	Vapour	173	170	58.4	0.0	0.0	0.0	52.6	5.8
40	Liquid	173	170	0.8	0.0	0.0	0.0	0.3	0.5

Abbreviations

kmol/h: kilomoles per hour; kPa: kilopascals (absolute); °C: degrees Celsius
HP / MP / LP steam: high- / medium- / low-pressure steam
Mixed: vapour-liquid mixture
Stream numbers in diamonds; continuation streams keep their number on both sheets

Toluene hydrodealkylation (HDA) plant — PFD-202	
Separation section | continued on PFD-201	
Sample process flow diagram; stream values are sample values	
Originator: Claude Code | 2026-10-24	Revision 1 | Sheet 2 of 2